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A gamified interactive educational tool to support algebraic thinking: reducing reversal errors in comparative word problem solving

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Abstract

This study evaluates the effectiveness of a gamified interactive educational tool (G-IET) in addressing reversal errors in algebra, a persistent misinterpretation of relational language in comparative word problems. The G-IET was designed and developed within a didactic framework specifically aimed at developing algebraic learning, providing a concrete visual learning approach and immediate feedback with the goal of enhancing students' understanding of equality and symbolic representation in algebraic contexts. A longitudinal design was employed with students who initially committed reversal errors. Participants engaged with the G-IET across eight sessions. Quantitative data were collected via pre- and post-tests involving algebraic translation tasks under both timed and untimed conditions. Qualitative data were obtained through open-ended responses and emotional engagement surveys grounded in flow theory. Results indicated substantial improvements in students' algebraic performance (large effect): an 87% reduction in reversal errors was achieved, accompanied by substantial decreases in operator errors (87.5%) and a complete elimination of hybrid errors (100%). Qualitative analyses revealed enhanced comprehension of problem statements, improved equation formulation, and greater self-regulation in learning strategies. Students highlighted the visual design, progressive structure, and motivational aspects of the tool as beneficial for learning. Emotional engagement remained consistently high throughout the learning experience, contributing to sustained attention and motivation. In conclusion, the G-IET demonstrated strong pedagogical effectiveness by significantly reducing reversal errors and supporting the development of conceptual understanding in algebra. These findings emphasize the importance of integrating interactive, visually supported environments in mathematics instruction to target and remediate specific misconceptions while maintaining student engagement.

Keywords: Reversal error, Comparative word problem-solving, Algebraic thinking, Gamified interactive educational tool, Flow, Human–computer-interaction

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1 Introduction

Problem solving is a core component of mathematics education (National Council of Teachers of Mathematics, 2000). Historically, word problem solving has held a prominent place in both primary and secondary education, as well as in everyday life (Halmos, 1980). Among mathematical problem types, *comparative word problems* play a particularly significant role because they require students to coordinate linguistic, arithmetic, and relational information simultaneously. These problems are essential in the transition from arithmetical to algebraic reasoning, where students must move from concrete computation to expressing relationships between quantities symbolically (Kieran, 2007; Levin & Walkoe, 2022).

Developing algebraic thinking has therefore become a central goal of mathematics education worldwide. Algebraic thinking goes beyond the manipulation of symbols: it involves recognizing patterns, identifying functional relationships, and expressing generalizations through mathematical representations (Radford, 2000). This kind of reasoning enables students to move from procedural arithmetic to structural understanding, seeing mathematics as a system of relationships rather than a collection of isolated calculations (Carraher et al., 2008). Research consistently shows that the early development of algebraic thinking enhances students' capacity for modeling, abstraction, and logical reasoning, forming a foundation for higher-order mathematical competencies (Blanton et al., 2011; Kieran, 2022). In this sense, promoting algebraic thinking from an early stage is not only a curricular priority but also a key factor in mathematical literacy and lifelong learning.

The significance of word-problem solving for mathematical literacy has been widely recognized in international assessments. As a result, such problems were included as one of the three evaluation criteria in the 2012 Programme for International Student Assessment (PISA) (Organisation for Economic Co-operation and Development; OECD, 2014). Since PISA 2003 (OECD, 2005), which introduced a separate problem-solving module to assess students' ability to apply mathematical knowledge in non-routine contexts, significant progress has been made in the specificity and refinement of assessment frameworks. Notably, PISA 2012 incorporated new methodologies that emphasized student interaction with the problem context. That year marked the beginning of formal assessment of students' problem-solving competence. In the context of algebraic problems, the subdomain "change and relationships" received particular attention, with the OECD average score reaching 494 points. However, the most recent PISA report (2022) (OECD, 2023) revealed a concerning decline in problem-solving outcomes. Although multiple factors may contribute to this trend, it is important to note that part of this decline may be associated with the educational disruptions caused by the COVID-19 pandemic. In the "mathematical reasoning" subdomain, the OECD average dropped to 473 points, and to 470 points in "change and relationships." These results indicate a decreasing ability among students to engage with complex mathematical situations and to select, compare, and evaluate appropriate problem-solving strategies.

Beyond cognitive skills, affective and emotional factors are increasingly recognized as crucial for successful mathematical learning. Recent research highlights that students' emotions, motivation, and engagement strongly influence their persistence, strategy use, and achievement in mathematics (Pekrun et al., 2023; Schukajlow et al., 2023).

Furthermore, *flow*, the state of deep, focused immersion in a learning activity, has been shown to enhance both cognitive and affective aspects of learning (Csikszentmihalyi, 1990; Jinmin & Qi, 2023).

The current Spanish secondary education curriculum (MEFP, 2022) aligns with this view, emphasizing problem solving and socio-emotional competencies as core components of mathematical learning. It highlights how developing self-regulation, motivation, and positive attitudes toward mathematics contributes to improving academic performance and reducing affective barriers to learning.

In this context, digital tools and interactive learning environments have emerged as promising resources for addressing both the cognitive challenges of mathematical reasoning and the affective challenges of engagement and motivation. Studies on interactive and gamified learning environments demonstrate that immediate feedback, progressive challenge, and visual representations can support conceptual understanding while maintaining learner engagement (Alsadoon, et al., 2022; Smirani & Yamani, 2024). Technology-based interventions that combine these design principles can support the development of algebraic reasoning while promoting positive emotional experiences that sustain learning.

Building on this evidence, the present study explores how a gamified interactive educational tool (G-IET) can enhance algebraic learning in comparative word-problem solving contexts, integrating cognitive and affective dimensions within a structured digital environment.

1.1 Comparative word problems and the reversal error

A significant line of research in arithmetic-algebraic word problem solving focuses on identifying the cognitive processes involved from the recognition of a mathematical relationship in a problem to its representation in algebraic form. From a cognitive perspective, solving arithmetic-algebraic word problems involves two phases, an analytical process followed by a synthetic process (Cerdán, 2008; Kalmykova, 1975), that is: (a) the analytical phase reduces the word problem statement into mathematical information by identifying and connecting quantities, progressing from the unknown to the known; and (b) the synthetic phase uses this structure to express unknown quantities in mathematical language based on previously established expressions. For example, when solving a problem such as "There are three more apples than pears, and together there are thirteen fruits. How many pears and apples are there?", the analytical phase involves identifying the quantities and their relationships in order to determine the steps needed to solve the problem. A student using analysis would think: "I don't know how many pears or apples there are, but I know that together they add up to 13 (apples + pears = 13) and that there are more apples, exactly three more than pears (apples = pears + 3)." The synthetic phase then consists of relating both analysis and representing them together algebraically (e.g., $x + (x + 3) = 13$) and assigning quantities to x to find the answer.

Ideally, arithmetic-algebraic problem solving could be viewed as two sequential and independent stages: a complete analysis, followed by synthesis. However, performing a complete analysis before synthesis is often challenging for students (Bogolyubov, 1972). In fact, it has been observed that when students solve problems using both arithmetic and algebraic methods, they tend to engage in repeated cycles of analysis and synthesis

rather than performing a thorough, comprehensive analysis. While this partial approach does not significantly impact performance in arithmetic problem-solving, it often leads to incorrect equations in algebraic contexts. For instance, when presented with the statement "There are three more apples than pears," many students incorrectly represent the relationship as $3 + apples = pears$, revealing a lack of understanding and analytical processing of the relational structure. This illustrates how insufficient analysis prior to synthesis can lead to errors in algebraic problem solving, in this case, the well-known reversal error, which serves as one example among several possible ones.

The reversal error occurs when students correctly identify the data in additive or multiplicative comparison word problems but invert the comparative relationship when translating it into algebra, switching the variables in the equation (Clement, 1982). This error occurs when students identify relationships between quantities using conceptual schemas of additive and multiplicative comparison. Its name originates from tasks such as: "Write an equation using the variables S and P to represent the following statement: There are six times as many students as professors at this university. Use S for the number of students and P for the number of professors" (Clement, 1982, p. 17). The correct equation is $S = 6 \cdot P$, but many students mistakenly write $P = 6 \cdot S$, reversing the intended relationship. The error derives its name from the reversal of the variables P and S, which are placed in positions inverse to their correct semantic roles in the equation.

A considerable body of research (Clement, 1982; Cooper, 1986; Fisher, 1988; Fisher et al., 2011; González-Calero et al., 2015, 2020; Rosnick, 1981; Soneira et al., 2018; Ventura-Campos et al., 2022a, b; Wollman, 1983) has investigated this error since it was first described by Kaput and Clement (1979) in a letter to the Journal of Children's Mathematical Behavior:

> *"...we have asked a variety of subjects to write an algebraic equation, using S for the number of students and P for the number of professors, that is equivalent to the following statement: 'At a certain university, there are six times as many students as there are professors.' Upwards of 25%–30% of freshman engineering students write $6S = P$ instead of $6P = S$." (p. 208)*

The reversal error has been identified across different educational levels, appearing among high school students at the early stages of formal algebra learning (MacGregor & Stacey, 1993) as well as among university students (Clement, 1982; Cohen & Kanim, 2005; Fisher et al., 2011; González-Calero, et al., 2015; Soneira, et al., 2018; Ventura-Campos et al., 2022a, b).

According to Clement et al., (1982), one major source of reversal errors is the word-order matching strategy, which is syntactic in nature. This model proposes that students translate verbal statements into equations by mimicking the linear order of words, rather than constructing a coherent mental model of the situation. For example, when reading "There are six times as many students as professors," a student applying this strategy might directly map the phrase into $6 \cdot S = P$, misinterpreting the structure as "six times students equals professors."

A second explanation is the static comparison model, which stems from semantic processing. Here, students mentally visualize the situation, e.g., one professor for every six students, and attempt to represent this image algebraically. Instead of analyzing the

mathematical relationship, they construct a symbolic shortcut: $1{\cdot}P = 6{\cdot}\mathrm{S}$, which they simplify to $P = 6{\cdot}\mathrm{S}$, thus making the same reversal error (Clement et al., 1982).

As far as we know, all instances of the reversal error reported in the literature have been designed and implemented as assessment instruments intended primarily for data collection by researchers. Their main objective has been to determine which explanatory model, static comparison or word-order matching, is more predominant among students, and how the linguistic formulation of the problems might reduce the incidence of the error. Very few studies have implemented instructional interventions to eliminate the reversal error. To date, and based on the current state of knowledge, only one study has addressed this issue within an interactive environment that transcends the traditional pen-and-paper approach. This study involved the development and use of POETIC (Purdue Optimization Modeling Education Tool—Interactive Equation Component), designed by Kim et al., (2014) and deployed on an interactive web-based platform. The study included 200 participants aged 18 to 65, from which 47% had completed higher education. Participants solved eight problems during a single session limited to 28 min. The intervention employed strategies such as graphical visualization of equations and variable substitution, aimed at helping learners visually represent algebraic relationships. Their findings indicated that the combined use of *visual representation* and *substitution strategies* can contribute to reducing the occurrence of reversal error. Nonetheless, their study failed to show significant advancement in fully addressing and eliminating the reversal error.

In addition to the limited progress observed in eliminating the reversal error, the study did not systematically differentiate whether participants' mistakes stemmed specifically from reversal errors, other difficulties in equation construction, or comprehension issues. The study also did not consider variables that the literature has identified as relevant to the occurrence of the reversal error, such as the presence of syntactic obstacles in the problem statements, the conceptual schemas of additive or multiplicative comparison (with all items in this case being of the multiplicative type), nor the types of magnitudes involved. Previous research (Soneira et al., 2018) has demonstrated that these variables significantly influence the prevalence of the reversal error.

As noted by Kim et al., (2014) and other studies (Ventura-Campos et al., 2022a, b; Author), traditional pedagogical approaches centered on abstraction have proven ineffective in addressing this type of error. This underscores the need to adopt more concrete and visually grounded instructional strategies, such as those initially proposed in the POETIC program, that enable students to comprehend the proportional relationship between the variables represented in an equation. According to Li et al., (2020), developing methods that enhance positive emotions in the teaching–learning process is relevant since they can significantly impact critical cognitive processes involved in learning, ultimately leading to improved academic achievement. Therefore, it is essential to consider emotional factors when designing and implementing new learning processes.

1.2 Emotions and mathematical learning

Learning is a complex process involving a network of neurobiological interactions across various regions of the brain. Research has demonstrated that learning is not solely determined by intellectual ability, but is also deeply influenced by emotional and motivational

factors (Bruner, 1996; Li et al., 2020). Cognitive processes such as memory, attention, emotions, motivation, and language are fundamental to learning and play a crucial role in knowledge acquisition, skill development, and the formation of attitudes (Li et al., 2020).

Although mathematics has traditionally been associated with rational and analytical thinking, it is now well established that students' perceptions of mathematics and their emotional responses to it have a significant impact on their performance (Wen & Dubé, 2022; Author). However, such affective dimensions are often overlooked in educational contexts. Various studies have shown that students with higher engagement levels tend to display greater enthusiasm, adopt more effective learning strategies, participate more frequently in collaborative activities (Guerra et al., 2019), and report higher satisfaction with their educational experiences (Pena et al., 2012).

In this regard, neuroeducation offers valuable insights by integrating cognitive enhancement strategies (e.g., attention, memory, reasoning) with emotional and motivational aspects of learning. This holistic approach helps students overcome common learning barriers and fosters deeper, more meaningful engagement with mathematical content. Numerous studies have demonstrated that learning is significantly enhanced when content elicits an emotional connection with students (Damasio, 2010; Mora, 2021). Positive emotions, in particular, play a critical role in enhancing focus, stimulating curiosity, and reinforcing motivation, all of which are essential for achieving deep, long-lasting understanding (Mora, 2021).

From this perspective, it is essential to design pedagogical strategies that are aligned with how the brain learns, promoting a balanced integration of cognitive and emotional processes in order to facilitate optimal learning experiences, including what is known as *flow.*

The concept of *flow* was introduced by Csikszentmihalyi (1975) as a state of optimal experience in which individuals are deeply engaged in an activity they find both enjoyable and absorbing. Flow refers to a mental state wherein individuals become fully immersed in a task, to the extent that they lose awareness of their own performance. In this state, perceptions, emotions, and actions are experienced positively and harmoniously, enabling the attainment of a high level of performance that occurs almost automatically and without conscious effort.

Students who experience flow in educational settings tend to achieve better academic outcomes, sometimes even outperforming those with higher baseline academic skills (Whalen, 1998). Flow not only enhances performance but also increases motivation and commitment to a level that surpasses what is typically observed in activities perceived as merely pleasant or entertaining.

1.3 Active methodologies: gamification in mathematics education

New teaching and learning methodologies should be both innovative and based on rigorous scientific standards. Consequently, it is essential to identify approaches that effectively enhance student motivation, thereby improving learning outcomes. In recent years, there has been a growing emphasis on assigning students a more active role in the learning process, while redefining teachers as guides and mentors. Accordingly, active learning methodologies have the potential to support this objective. These methodologies enhance cooperation, motivation, and learner autonomy, and include approaches

such as cooperative learning, flipped classrooms, gamification, problem-based learning, real-life applicable projects, and service-learning (Diaz-Palencia et al., 2023).

Play has a crucial role in both emotional and cognitive development, as it stimulates dimensions of self-awareness, knowledge, and action (Valverde, 2012). Consequently, its integration as an educational strategy is highly valuable for promoting skills such as problem-solving, collaboration, and communication.

In this regard, gamification involves the integration of fundamental game design elements into non-ludic contexts (Deterding et al., 2011). As noted by Gallego et al., (2014), this instructional strategy enhances the teaching–learning process by leveraging game dynamics, thereby promoting increased engagement and motivation toward the learning content. The primary aim of gamification is to deliver learning experiences that are engaging, motivating, and meaningful by incorporating game rules and mechanics within virtual environments or other educational platforms (NMC, 2013).

The incorporation of Information and Communication Technologies (ICT) as tools for managing and processing information has significantly transformed traditional educational approaches. ICT has paved the way for more meaningful, constructivist learning models (Guerrero, 2014). As highlighted by Sosa et al., (2010), when properly integrated, ICT significantly enhances the dynamics of teaching and learning. Aguilar (2012) further notes that ICT not only enriches the educational value of content, but also helps to transcend the limitations of traditional pedagogical models, promoting a more open and innovative form of education.

This study is structured around two interconnected aims. First, it evaluates the effectiveness of the G-IET (Author et al., 2023), specifically designed and developed to address the reversal error when players solve algebraic comparative word problems. Second, it examines the didactic approach embedded in the tool, based on visual representations, structured levels, and immediate feedback, to determine its potential to enhance students' understanding of algebraic relationships, particularly equality and relational structure in comparative word problems, and, consequently, to improve or eliminate the occurrence of reversal errors. Furthermore, it investigates students' emotional responses to their learning experience while interacting with the G-IET, alongside their evaluation of the instructional effectiveness of the tool. Additionally, this study also examines whether the implementation of gamified elements within the tool could not only improve conceptual understanding of reversal error problems but also positively affect students' emotional engagement during the learning process.

2 Methodology

2.1 Participants

The study comprised a sample of 52 university students aged 18 to 26 years, including 32 females. Participants were drawn from diverse academic programs spanning the sciences, health sciences, social sciences, and humanities. This population was intentionally selected, as the aim was to evaluate the G-IET with students who had already acquired formal algebraic competencies during secondary education. Our study specifically focused on identifying individuals who, despite having completed secondary education, continued to exhibit reversal errors, an indication that they had not yet developed a deep conceptual understanding of comparative verbal statements. Participants were recruited

through informational posters placed across the university campus inviting students to take part in a study on educational tools and learning processes. Participation was voluntary.

The study employed a pretest–posttest quasi-experimental design, a methodology commonly used in educational research when participants are selected for their accessibility and relevance to the study's aims (Gray and Groove, 2017). A purposive, criterion-based sampling strategy (Kelly, 2010; Palinkas et al., 2015) was applied to recruit students aligned with the specific objectives of the study. All students first completed a pretest, which was used to determine whether they exhibited the reversal error. Only those who showed this error were included in the intervention phase. After fully interacting with the G-IET, these selected participants completed a posttest.

The research protocol received approval from the Ethical Committee of University Jaume I (information anonymized) and was conducted in full compliance with applicable guidelines and regulations. Informed written consent was obtained from all participants before their involvement. Furthermore, participants received economic compensation for their participation, including an additional 10€ bonus for those who exceeded 90% accuracy from Level 4 onward in the G-IET (see Sect. 2.3).

2.2 Reversal error detection tasks

To identify participants suitable for the study, two tasks were implemented to detect reversal errors in the resolution of comparative word problems. The first task, known as the Time-Limited Reversal Error Task (*RE-task TL*), required participants to decide within a 3-s time limit, whether the equation presented for a given problem statement was correct or incorrect. The equations shown could be correct or include a reversal error. This constraint was designed to minimize reliance on validation strategies and increase the likelihood of observing reversal errors stemming from misinterpretation of mathematical relationships.

The second task, the No Time Limit Reversal Error Task (*RE-task NoTL*), allowed participants unlimited time to write the equation that best represented the problem statement. This setup enabled researchers to assess whether reversal errors persisted when participants were free to reflect, offering insights into their conceptual understanding and ability to translate comparative word problems into algebraic form.

2.2.1 RE-task TL

In the time-constrained task, participants completed a computerized assessment developed to detect reversal errors (Author). Visual stimuli were presented using E-Prime software (version 3, Psychology Software Tools, Pittsburgh, PA), which also recorded response accuracy and reaction times.

This task included two conditions, reversal-prone and reversal-free, and consisted of 48 trials in total (24 per condition), organized into eight thematic blocks based on different problem types. Each block contained both reversal-prone and reversal-free statements, with two items per condition. In the reversal-prone condition, the problem statements were likely to elicit reversal errors, whereas in the reversal-free condition, they were structured to avoid such errors. The block order was randomized and balanced for difficulty. Each trial followed a fixed sequence: a 1-s fixation point, an 8-s

display of the problem statement (in Spanish), and a 3-s window to indicate whether the presented equation was correct or incorrect. Responses were categorized as correct or reversal error. In half of the trials, the displayed equation accurately represented the statement; in the other half, it contained a reversal error. Participants indicated whether the equation was correct or incorrect by pressing the 'C' or 'I' key, respectively. The full task lasted 9 min and 36 s. Detailed timing and structure are shown in Fig. 1a.

Reversal-prone items involved discrete quantities and varied across three dimensions: (1) the type of relationship (additive or multiplicative) (González-Calero et al., 2015; MacGregor & Stacey, 1993; Soneira, et al., 2018); (2) increasing comparisons or decreasing comparisons In English, multiplicative comparisons are typically expressed using constructions such as "N times as many X as Y" (e.g., "There are six times as many students as professors"). The most literal equivalent in Spanish would be "N veces tantos X como Y." However, in common Spanish usage, multiplicative comparisons are more frequently expressed using the syntagms "N veces más que" and "N veces menos que" (Castro, 1995), whose direct, word-for-word translations into English would be "N times more than" and "N times less than," respectively. In relation to additive comparisons**,** we used the expressions "más que" ("more than") and "menos que" ("less than"); and (3) the presence or absence of contextual cues (Cohen & Kanim, 2005; González-Calero, et al., 2015; Soneira, et al., 2018; Wollman, 1983) that may facilitate the identification of the quantity that is typically greater.

Reversal-free items mirrored the structure of the reversal-prone ones but avoided reversal errors by presenting clear, unambiguous relationships. For example: "At an academy, the number of tutors (T) plus 9 equals the number of students (A)," with the correct equation being $T + 9 = A$. These items ensured that participants could accurately translate straightforward statements into equations. This design ensured that any observed difficulty could not be attributed to a lack of knowledge about constructing equality equations.

2.2.2 RE-task NoTL

In this second task, participants generated equations using a computer-based interface similar to that described by González-Calero et al., (2015). The task comprised 16 Spanish-language statements covering all combinations of four variables: (1) comparison type (additive vs. multiplicative), (2) comparison direction (increasing vs. decreasing), (3) presence of contextual clues, and (4) type of quantities (discrete or continuous). Participants were restricted to using either addition/subtraction or multiplication/division when formulating the equation. To facilitate this process, the interface included a "Quantities" box containing both the variables and the numerical value needed for each item, alongside calculator-style buttons for the relevant mathematical operators (see Fig. 1b). It is important to note that response time was not restricted. Participant responses were classified as correct, reversal errors, or other errors.

For instance, in the multiplicative example "There are nine times as many passengers as stewardesses," with P representing passengers and S stewardesses, the correct equation is $P = 9 \cdot S$. Equivalent formulations (e.g., $S = P/9$; $P/S = 9$) were also accepted as correct. A reversal error would correspond to an equation such as $S = P \cdot 9$, or an equivalent formulation. All other formats were categorized as *other errors*. These *other errors* encompassed

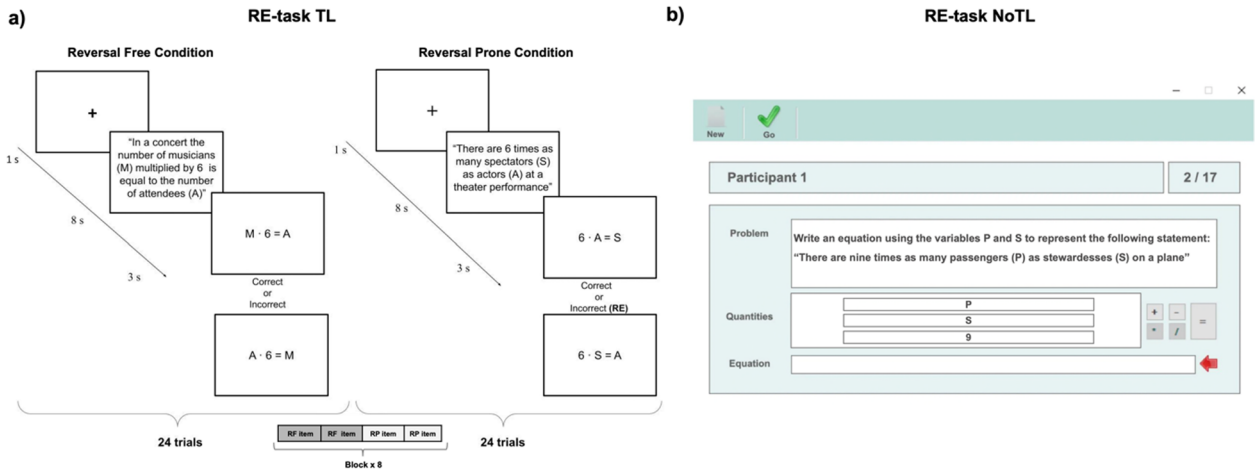


Fig. 1 Schematic representation of the experimental design of the RE-task TL and RE-task NoTL. *Note.* **a** RE-task TL: RF = Reversal Free Condition; RP = Reversal Prone Condition. Each block presenting the same item corresponds to one of the eight experimental statement combinations (e.g., Multiplicative-Increased-Contextual, Multiplicative-Increased-Noncontextual, Additive-Decreased-Contextual, etc.). **b** RE-task NoTL: Participants constructed the equation corresponding to the problem statement by clicking on variables (quantities) and operation signs (+, −, *,/, and =) located to the right of the quantities. The rectangle labeled "Equation" displayed the variables and operators they selected. Once completed, participants validated their equation by pressing the "Go" button

two main types. The first involved operator errors resulting from linguistic confusion with comparative expressions. For instance, participants often misinterpreted phrases such as "times more than" or "times less than" incorrectly associating them with additive instead of multiplicative operations (e.g., writing $P = S + 9$ instead of the correct $P = S \cdot 9$). Conversely, expressions like "more than" or "less than,"which imply addition or subtraction, were sometimes mistakenly linked to multiplicative operations. This confusion may stem from the fact that in Spanish, multiplicative expressions are literally translated as *"veces más que"* or *"veces menos que"* ("times more/less than"). The misunderstanding may also be reinforced by the way multiplication is taught in Spanish: while in English the operation is expressed as *"*N times M" (where N and M represent numerical factors), in Spanish it is taught as *"*N por M*"*. As a result, learners may overlook that the word *"veces"* ("times") itself denotes multiplication, which can lead to misinterpretations when reading expressions like *"veces más que"*. The second type of error was structural in nature and reflected a misunderstanding of how variables function. In these cases, participants constructed expressions such as $9/P = S$ instead of the correct $P/9 = S$, or in additive contexts, $9 - P = S$ instead of $P - 9 = S$. This difficulty may stem from different algebraic misconceptions documented in the literature (Tularam and Hassan, 2025). A central issue appears to be that some students do not yet understand that a variable represented by a letter can take any numerical value within a relational structure. Previous research has shown that learners often interpret variables as fixed labels, as specific objects, or as placeholders for a single expected value, rather than as quantities that may vary freely, sometimes assuming that the variable cannot exceed the numerical value given in the problem statement (Küchemann, 1980; MacGregor & Stacey, 1997). Students may also impose implicit restrictions on the values that variables can assume, preferring expressions that avoid negative or fractional outcomes and that align more closely with familiar arithmetic patterns (MacGregor & Stacey, 1997). Notably, these errors could occur independently or in conjunction with reversal and operator errors.

2.3 Gamified interactive educational tool

Research studies in the literature highlights a growing diversity of digital environments to support mathematics learning, ranging from video games to serious games, simulations, and interactive learning tools (Clark, et al., 2016; De Freitas & Jarvis, 2006; Devlin, 2011; Gee, 2003). Traditional video games typically rely on narrative progression, player agency, and open-ended exploration, while serious games embed full game mechanics within an educational purpose (Michael & Chen, 2005). In contrast, interactive learning environments focus primarily on structured instructional sequences and immediate feedback to foster conceptual understanding (Clark & Mayer, 2023). The tool developed for this study does not incorporate the narrative structure, fictional setting, or free-form decision-making characteristic of video games and serious games. Instead, it provides a guided learning experience built around visual representations, progressive task complexity, and feedback. At the same time, it includes two game-like elements, structured levels and a coin-reward system from level 4 onwards, which align with the notion of gamification as the integration of selected game elements in non-game contexts to enhance engagement (Alsadoon, et al., 2022; Deterding et al., 2011). For these reasons, the tool is more accurately described as a Gamified Interactive Educational Tool (or

G-IET) designed specifically to support the development of algebraic thinking by helping students understand equality and relational structure in comparative word problems, while maintaining a lightweight motivational layer. In this sense, the tool can be considered a type of gamified Interactive Learning Environment or ILE (Koç & Kanadlı, 2025), characterised by structured instructional sequences with light gamified reinforcement.

This G-IET (Author) was designed to enable students to gain a deep understanding of the problem statement and its correct translation into an algebraic equation (see Fig. 2, not shown to preserve anonymity). The initial prototype (Author) was extended to become a digital training tool with different levels of difficulty aimed at helping users overcome reversal errors. It has been built using Unity3D version 2021.3.12f1 and the application is compiled for Android platforms. It runs smoothly on mobile devices or tablets with Android version 5.1 (Lollipop, API level 22) or newer. The G-IET recorded each participant's constructed equations along with their corresponding response times in a database stored in Google Firebase. Designed as a G-IET, "Name App[1]" (not shown to preserve anonymity) is intended for secondary school students. The objective is to provide students with tools to develop further their skills to overcome reversal errors before transitioning to higher education or entering the workforce.

Our G-IET adopts the approach of visual representation and substitution strategies implemented in the POETIC program (Kim et al., 2014), while also improving both the visual design and the learning environment. These enhancements are based on the observation that POETIC's visual representations were not sufficiently clear or engaging for students at the intended educational level (final year of primary and secondary education). In addition, the G-IET is designed to support step-by-step learning and incorporates other relevant features, such as the types of problem statements identified by Soneira et al., (2018), inverse task aimed at fostering reversible thinking (Piaget, 1979), mechanisms to address *other errors*, and an affective evaluation of the learning process during interaction with the G-IET. These features are detailed below:

The G-IET consists of eight levels and each level includes a set of 10 exercises. The narrative of these levels is to follow a path on an island, analogically representing the learning path that a student experiences while learning how to avoid committing reversal errors. At each stop in the path, the learner needs to complete 10 exercises before progressing to the next step towards the final goal. These exercises are dynamically selected and retrieved from a database hosted on Firebase Firestore. Each exercise is structured into several interactive phases: (1) *Understanding the Problem,* where the user is presented with a written word problem; (2) *Formulating an equation,* using an on-screen calculator, the user creates an algebraic equation that models the situation described; (3) *Solving and Feedback*, if a reversal error is detected, the tool initially applies a correction based on arithmetic validation, using the substitution strategy. In this phase, numerical values are inserted into the variables of the equation proposed by the player, allowing them to verify whether the resulting expression accurately reflects the intended comparative relationship. When a reversal error is present, the substitution reveals that the equation does not satisfy the equality. In that case, the tool moves to a new screen to deepen the player's understanding of the comparative statement. At this

[1] Registered Software at University Jaume I, information anonymized.

stage, a second corrective approach is used, applying the visual strategy to represent the statement. The tool presents a graphical visualization that illustrates the relative magnitudes of the two variables described in the problem, using container sizes as physical metaphors. This representation helps the player understand the logical structure of the comparison (who is greater or smaller, or how many [times] more or less) and reinforces an accurate relational interpretation. Based on this visual representation, the tool invites the participant to reflect on how the magnitude of one of the two containers could be transformed to satisfy the equality, presenting the following question: *Which variable would you multiply/divide/add/subtract by N so that the equality holds, that is, so that both containers reach the same size?* Here, "variable" refers to one of the two containers, whose size represents the magnitude conveyed in the comparison statement (see Fig. 2x) (e.g., "there are six times as many students as professors"; in this case, the question was: *Which variable would you multiply by 6 so that the equality holds, that is, so that both containers reach the same size?*). Starting from Level 4, the visual strategy changes the way it guides the participant when the reversal error is made: the relative size of the containers representing the statement is no longer provided explicitly. Instead, two identical containers are shown, and the participant must determine which one should be larger or smaller according to the problem statement. Once the sizes are selected, the participant is asked to choose the operator that best represents the described relationship and decide to which variable it should be applied. In this way, from this level onward, the tool becomes less directive and requires greater reflection from the participant. In contrast, if the equation does not contain reversal errors, only the substitution strategy is applied, with the goal of consolidating the understanding of the relationship between the variables. To this end, specific numerical values are substituted into the variables to verify the equality between the related quantities.

The problem statements within the G-IET have been created to address varying levels of cognitive complexity. Syntactically, the statements encompass both those with and without structural obstruction, incorporate different types of comparison (additive and multiplicative), and involve diverse magnitudes, adhering to the criteria established by Soneira et al., (2018). From the perspective of semantic content and the development of reversible thinking, the tool includes an inverse task in which users are asked

(See figure on next page.)

Fig. 2 Gameplay sequence of the G-IET across levels 1–8. Interface texts in the figure are presented in Spanish; **a** literal English translation is provided here. **b** The following statements describe situations where a comparison between two quantities is made. Read and try to understand the meaning of the following statement: Levels 1–3: *In a civil service exam, there are 7 times more candidates than positions.* Levels 4–8: *In a library, there are 8 times fewer people than books.* (c) Write the equation using the variable *(O)* for the number of candidates and *(P)* for the number of positions. (d) This is your equation. If your variables now represent *O = orchids* and *P = palm trees*, write in your own words a new statement that gives meaning to your equation. (e.1) Which is greater, the number of candidates or the number of positions? (e.2) and (h) Let us see what happens if we substitute the variables with quantities. Considering the statement, suppose that *O = 7* and *P = 1.*//This is your equation for the comparison problem posed; let us see whether the equality holds.//*Does the equality hold?* (f) Levels 1–3: Which variable would you multiply by *7* so that the equality holds (i.e., so that the size of the two boxes is equal)? Levels 4–8: (1) Select the variable that, according to the statement, is *8 times fewer*. (2) Select the variable and choose the operator you want to apply to it, ensuring that it corresponds to the meaning of the statement. (g) Levels 1–3: Write the equation using *(O)* for the number of candidates and *(P)* for the number of positions. Levels 4–8: Write the equation using *(L)* for the number of books and *(P)* for the number of people. (i) *You are doing an incredible job! Keep it up!*

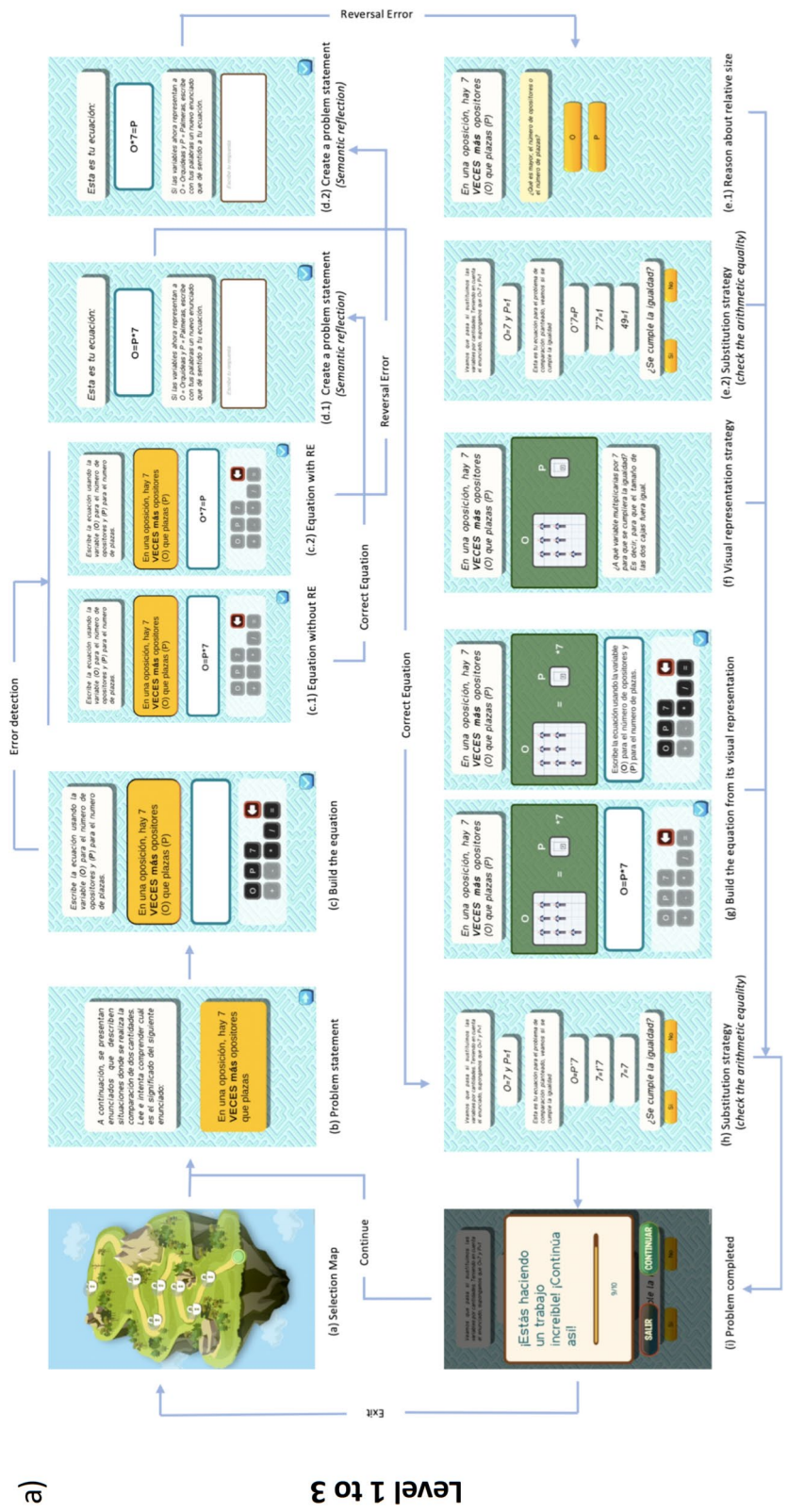


Fig. 2 (See legend on previous page.)

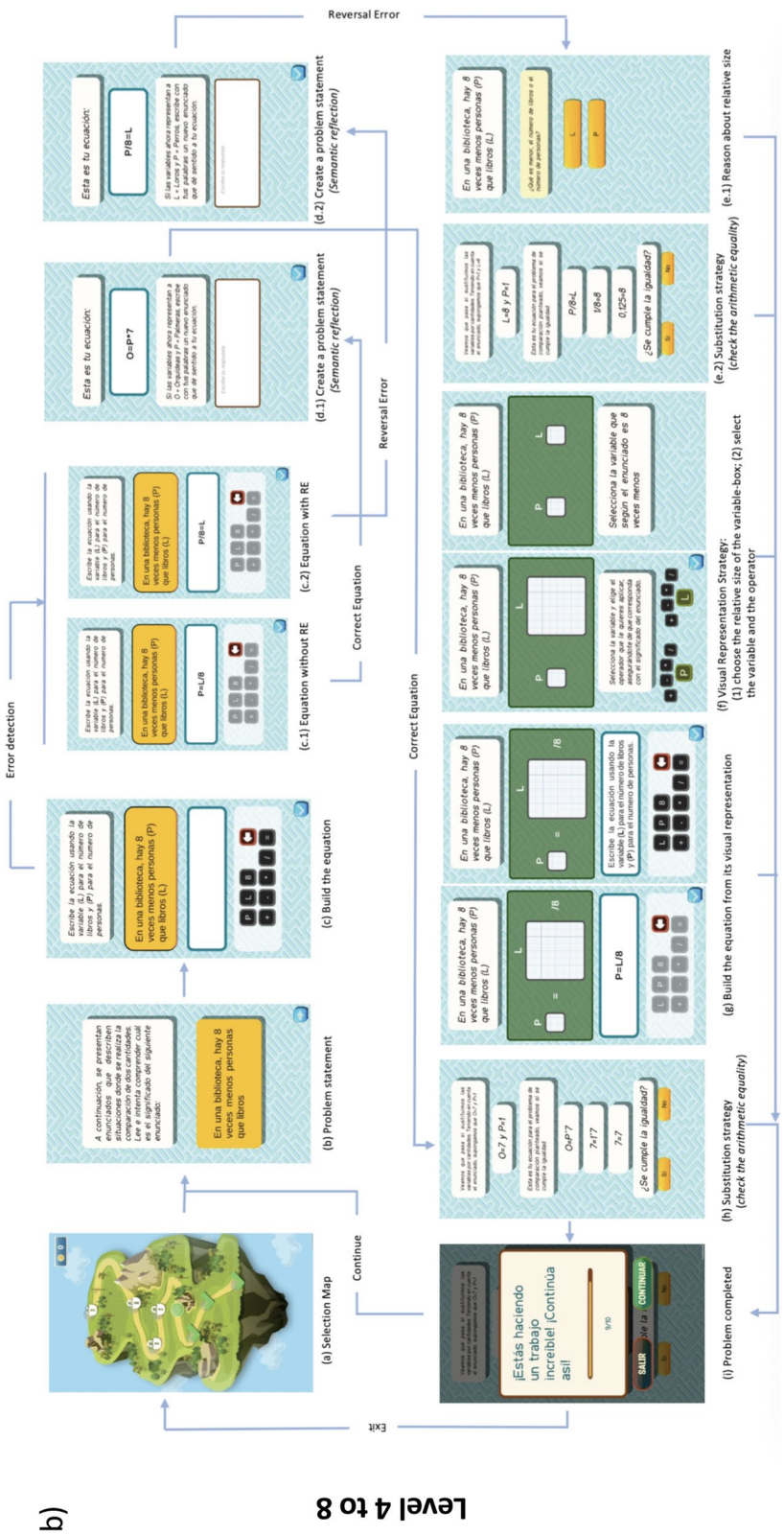


Fig. 2 continued

to construct a problem statement based on an equation they previously created. This activity encourages them to reverse the reasoning process and, in doing so, deepen their understanding of the underlying algebraic relationships.

The G-IET incorporates one of the most recent advances in the analysis of the reversal error within mathematical learning. Beyond Clement's (1982) two models, where the error arises from either a linear translation or a static comparison, Jankvist and Niss (2021) propose a combined syntactic and semantic explanation. They argue that the idiom "X times as many as" can pose difficulties for students who have not fully developed an understanding of the concept embedded in this linguistic structure, potentially leading to operator errors. To support comprehension, when an operator error is detected, whether related to expressions such as "times as many as," "times fewer than," "more than," or "less than", the tool displays an explanatory message. This message uses concrete examples to clarify the meaning of these expressions and recommends the appropriate mathematical operator for each case.

The G-IET also addresses another error highlighted in the RE-task NoTL, which occurs when translating the problem statement into an equation involves placing the number as a divisor of, or subtracting from, the variable. In such cases, an error message alerts participants that the variable may be greater than the number and prompts them to revise the equation.

The G-IET interface includes a map-style level selection menu that allows players to track their progress: completed levels are marked, and available levels can be accessed by tapping circular buttons. After solving an exercise, users can view their progress on a visual progress bar. Furthermore, the tool features a clear and well-organized visual design, free of distracting elements, along with an intuitive interface that includes only the options necessary to progress through the tasks. This approach facilitates its use by secondary school students and helps them focus on the mathematical content without cognitive overload.

From level 4, participants were presented with an on-screen message informing them that they were about to face a new challenge. From that point onward, each correct answer was rewarded with one virtual coin, which accumulated on a visible counter within the G-IET interface. Participants could earn up to 50 coins by the end of the training. This motivational reinforcement was introduced in response to findings from pilot studies, which showed that by level 3, most participants had already acquired the core learning objective and began to show signs of reduced concentration and engagement. The coin system was therefore implemented as an encouraging strategy to maintain/increase attention and effort across the remaining levels. In the context of this study, which involved university students, the coin system was linked to a performance-based incentive: those who collected 45 or more virtual coins received an additional 10€ economic compensation. Although a monetary reward was used here, the coin-based system is inherently adaptable and can serve as a motivational tool in secondary education settings, where the incentive may simply be the challenge of outperforming peers or collecting as many coins as possible.

Additionally, the G-IET design incorporates an assessment of the affective dimension, conducted within the tool environment at the end of each level and upon completion of

the entire G-IET (see the variables and instruments section for a detailed description of the emotional variables).

2.4 Training sessions

Participants who were classified as exhibiting reversal errors, based on their performance on the RE-task TL and RE-task NoTL administered during the pre-intervention assessment, were selected to undergo training with the G-IET. They engaged in structured training over a three-week period, attending two to three supervised laboratory sessions per week using university-provided tablets. In most cases, one level was completed per session; however, due to scheduling constraints, the final levels were occasionally grouped. Each level required approximately 10 to 20 min to complete, depending on individual progress. After completing all eight levels of the intervention, participants then completed the posttest assessment, which was administered two to four days later, depending on their availability.

2.5 Variables and instruments

In each of the eight levels in the G-IET, the design incorporates emotional variables by evaluating participants' levels of enjoyment, concentration, and relaxation at the end of each level, using a Likert scale ranging from 0 (lowest) to 6 (highest). This aimed to capture their emotional engagement throughout the training process. Additionally, after completing all eight levels, the G-IET provides a final emotional evaluation. Participants were asked to provide an overall evaluation of their flow experience through five items. Each item was rated on a 6-point Likert scale ranging from 0 (not at all) to 6 (completely), capturing core dimensions of flow: time distortion, loss of self-awareness, positive affect, focused attention, and enthusiasm (Csikszentmihalyi, 1990) during interaction with the G-IET. These items included: (1) time seemed to fly while I was playing; (2) I became unaware of my surroundings; (3) I felt good while playing; (4) I was deeply focused while playing; and (5) I felt excited while playing the game. It is important to note that this evaluation was designed to capture participants' retrospective self-reported perception of their flow experience throughout their interaction with the G-IET over the 2–3 week intervention period.

A qualitative approach was adopted to explore and understand participants' perceptions of their learning experience with the G-IET as a didactic tool. The self-report questionnaire was administered via the Qualtrics platform immediately after participants completed the 8th level in the G-IET and finished the entire learning process, ensuring that students were able to reflect on how they believed they solved comparative word problems before instruction, as well as on the learning gains they perceived after working with the G-IET. This questionnaire included five open-ended questions, each designed to explore a distinct dimension of the educational experience:

1. *Pre-intervention error patterns*. Participants were asked to describe the types of mistakes they typically made when translating comparative word problems into algebraic equations prior to using the G-IET (e.g., "Before playing the game, what kinds of errors do you think you made when converting a statement into an equation?").

2. *Pre-intervention strategies.* This question explored the reasoning or strategies learners employed to construct equations before the intervention (e.g., "Before playing the game, what strategy or thought process did you use to write the equation?").
3. *Post-intervention comprehension.* Participants reflected on whether the game enhanced their understanding of word problem statements and their translation into algebraic form (e.g., "After playing the game: do you think the game has helped you understand the problems presented in these types of statements and translate them into equations?").
4. *Engaging with the G-IET feature*. Participants identified the elements of the tool they found most appealing or enjoyable (e.g., "What did you like the most about the video game?").
5. *Overall pedagogical utility*., Participants provided their general evaluation of the tool's usefulness as a teaching resource (e.g., "In your opinion, how effective is the video game as an educational tool? Would you recommend it?").

2.6 Data analysis

2.6.1 RE-tasks response analysis

The design of the two tasks enabled the classification of participants according to the presence or absence of the reversal error in their responses. In the RE-task TL (with time limit), participants were classified as showing reversal errors if they answered 50% or fewer of the items correctly. This threshold was used because the task involved judging whether given equations were correct or incorrect, and a 50% accuracy rate could result from random guessing. Therefore, scoring at or below this level was taken as an indicator of conceptual misunderstanding. In contrast, in the RE-task NoTL (without time limit), participants were classified as not showing reversal errors if they scored 65% or higher. This criterion was used to ensure that participants included in the group who did not make reversal errors had sufficiently acquired the necessary algebraic competence.

Based on these criteria, participants were assigned to three groups: (1) the *No-RE group*, consisting of participants who did not make reversal errors in either task; (2) the *RE group*, composed of participants who committed reversal errors in both tasks, indicating persistent conceptual difficulties; and (3) the *RE-COtime group* (time-COnditioned Reversal Error), including those who scored 50% or lower on the RE-task TL but 65% or higher on the RE-task NoTL. This group represents participants who exhibited reversal errors only under time constraints. This classification procedure made it possible to identify distinct performance profiles in solving comparative word problems, based on the presence of reversal errors and the role of time constraints.

For the purposes of the present study, only participants who initially made reversal errors (i.e., RE and RE-COtime groups) were included in the intervention analyses.

2.6.2 Longitudinal design: pre- and post-intervention assessment

In order to analyze the impact of the G-IET intervention on students' algebraic problem-solving abilities and emotional responses, both descriptive and inferential statistical

methods were employed. All analyses were performed using IBM SPSS Statistics (Version 29.0.2.0; IBM Corp., 2023) for Macintosh.

Descriptive statistics were initially computed to summarize the number of correct responses and errors made by participants in the RE-task TL and RE-task NoTL, both before and after the intervention. These descriptive indicators provided a first overview of participants' algebraic performance and error types before and after interacting with the G-IET.

To analyze participants' accuracy throughout the training, the total number of coins earned during interaction with the G-IET was recorded. As each correct response from level 4 onward was rewarded with a coin, this metric served as a cumulative indicator of performance. A descriptive analysis was conducted to examine the frequency distribution of total coins earned, allowing identification of performance patterns across participants. This analysis made it possible to determine how many participants reached or exceeded the threshold established for the performance-based incentive.

Before conducting inferential analyses, the normality of the data was assessed using the Shapiro–Wilk test. The assumption of normality was violated in several variables, including the number of correct responses on the RE-task NoTL at both the pre-intervention ($W = .864$, $p = .001$) and post-intervention phases ($W = .872$, $p = .002$), and in the RE-task TL post-test scores ($W = .783$, $p < .001$). The RE-task TL pre-test scores were normally distributed ($W = .962$, $p = .354$). Additionally, non-normal distributions were found in the average scores across the eight levels for enjoyment ($W = .846$, $p < .001$), relaxation ($W = .894$, $p = .006$), and concentration ($W = .962$, $p < .001$). Similarly, the emotional engagement variable derived from a principal component analysis (PCA) ($W = .895$, $p = .006$) and the perceived flow construct ($W = .902$, $p = .010$) also failed to meet the normality criterion.

Given these results, non-parametric statistical procedures were used throughout the analyses. To assess learning gains, Wilcoxon signed-rank tests were applied to compare pre- and post-intervention performance within each group for two dependent variables: the number of correct responses in the RE-task NoTL and the RE-task TL. These within-subject comparisons allowed for evaluating individual progress. Effect sizes (r) were calculated and interpreted according to Cohen's (1998) criteria: small ($\geq .10$), medium ($\geq .30$), and large ($\geq .50$).

To explore the evolution of emotional engagement throughout the G-IET, a Friedman test was conducted to detect significant changes in self-reported enjoyment, relaxation, and concentration across the eight game levels. Post hoc pairwise comparisons were performed using Wilcoxon signed-rank tests with Bonferroni-adjusted significance levels to account for multiple comparisons.

A principal component analysis (PCA) was conducted on the mean scores of enjoyment, relaxation, and concentration across levels to identify latent patterns of emotional engagement. The adequacy of the data was evaluated using the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's test of sphericity.

Perceived flow was measured at the end of the intervention using a five-item self-report questionnaire adapted from established flow scales. The internal consistency of the scale was assessed using Cronbach's alpha, resulting in a value of $\alpha = .88$, indicating excellent internal reliability for the five-item flow measure. Descriptive statistics (mean

and standard deviation) were calculated for each item and for the flow score, derived from the average of all five items, to summarize the participants' overall perception of flow throughout the gamified learning process. This analysis provides insight into the degree of immersion and enjoyment experienced by participants throughout the gamified learning process, particularly in relation to understanding comparative word problems and reducing reversal errors.

A Spearman's rank-order correlation analysis was conducted to examine the relationship between perceived flow and emotional engagement (PCA) during participants' interaction with the G-IET.

Finally, to explore whether emotional engagement and perceived flow differed between groups based on reversal error profiles, Mann–Whitney U tests were conducted to compare the PCA-based emotional engagement scores and perceived flow scores between the RE and RE-COtime groups. This analysis aimed to determine whether initial differences in reversal error patterns might influence affective and cognitive engagement during the G-IET intervention.

2.6.3 Qualitative data collection and thematic analysis

To analyze the qualitative data, a thematic content analysis was carried out to identify meaningful patterns and emergent categories in participants' responses. This approach enabled the systematic breakdown, organization, and in-depth interpretation of the information collected through the five open-ended questions, thereby facilitating a richer understanding of the perceptions, experiences, and reflections expressed.

The analytical process unfolded in several phases. Initially, an exploratory reading of the textual corpus was undertaken to gain familiarity with the overall content. This was followed by an initial coding phase, during which relevant units of meaning were identified. These units were subsequently grouped into thematic categories developed through a mixed approach: inductively, based on the data emerging from the participants' responses, and deductively, informed by the study's objectives and the theoretical framework underpinning this research (Saldaña, 2009).

To enhance the validity of the analysis, cross-checking procedures among researchers were applied, which helped to minimize individual bias and strengthen the consistency of the interpretations.

To ensure participant confidentiality and anonymity, an alphanumeric coding system was implemented. In this system, the letter "P" was used to designate each participant, followed by a number corresponding to their order of participation. For instance, the code P5 referred to the fifth participant in the study.

3 Results

The data obtained from the RE-task TL and RE-task NoTL before the intervention with the G-IET were used to classify participants into three groups based on their initial performance: the Non-RE group (22 participants, 12 female; $M = 20.9$, $SD = 2.01$), the RE group (13 participants, 11 female; $M = 21.85$, $SD = 2.51$), and the RE-COtime group (17 participants, 9 female; $M = 21.24$, $SD = 2.54$). For the purposes of the present study, only

participants who initially made reversal errors were selected to undergo the G-IET intervention, that is, those belonging to the RE and RE-COtime groups.

3.1 Longitudinal results: pre- and post-intervention performance

In the analysis of participants' responses, correct answers were defined as those that accurately reflected the mathematical relationship described in the comparative word problems. Figure 3 presents the mean number of correct responses and standard deviations for each group across both task formats (RE-task TL and RE-task NoTL), illustrating the improvements observed after the intervention. Incorrect responses were primarily attributed to reversal errors, where the structure of the equation inverted the variables in the comparative relationship. However, a portion of the incorrect answers was classified as "other errors," which included mistakes not directly related to variable reversal. These other errors typically stemmed from incorrect operator selection, often due to linguistic expressions or idiomatic phrases (e.g., misinterpreting "times more," "times less," "more than," or "less than"), or from applying subtraction or division to the number instead of the variable (e.g., 9/P = S). It is important to note that some of these "other errors" also involved a reversal component, making them hybrid errors. Therefore, although reversal errors were the most frequent cause of incorrect responses, a more nuanced understanding must account for these overlapping error types as part of students' broader algebraic difficulties. Notably, the G-IET also addressed these alternative errors by guiding students to correct them during equation construction. Figure 4 illustrates the relationship between reversal errors and other types of mistakes, showing that the tool helped eliminate not only reversal errors almost entirely but also significantly reduced other common algebraic errors among students.

To assess learning gains, non-parametric Wilcoxon signed-rank tests were performed to compare participants' performance before and after the intervention on both task formats. Results indicated statistically significant improvements in accuracy following the use of the G-IET. Specifically, in the RE-task NoTL (untimed format), correct responses significantly increased from pre- to post-intervention ($Z = -4.711$, $p < .001$), with a large effect size ($r = .86$). Similarly, in the RE-task TL (timed format), performance also improved significantly ($Z = -4.786$, $p < .001$), with a large effect size ($r = .87$). According to Cohen's conventions, effect sizes above.5 are interpreted as large, suggesting that the intervention had a strong impact on students' ability to correctly solve comparative algebraic word problems, particularly by reducing the occurrence of reversal errors.

The results of the non-parametric Friedman test showed that the emotional states of relaxation ($\chi^2(7) = 27.82$, $p < .001$) and concentration ($\chi^2(7) = 15.40$, $p = .031$) varied significantly across the eight levels of the G-IET, whereas enjoyment remained stable ($\chi^2(7) = 7.57$, $p = .372$). Although Bonferroni-corrected post hoc comparisons did not reveal significant differences between specific levels, uncorrected results suggest a meaningful pattern of change beginning at Level 6. Comparisons such as Level 1 vs. Level 6 ($\chi^2(7) = 2.77$, $p = .006$) and Level 2 vs. Level 6 ($\chi^2(7) = 2.32$, $p = .020$) indicate a notable increase in perceived relaxation in later stages of the game, potentially reflecting an emotional adaptation process as players gained mastery over the comparative word problems.

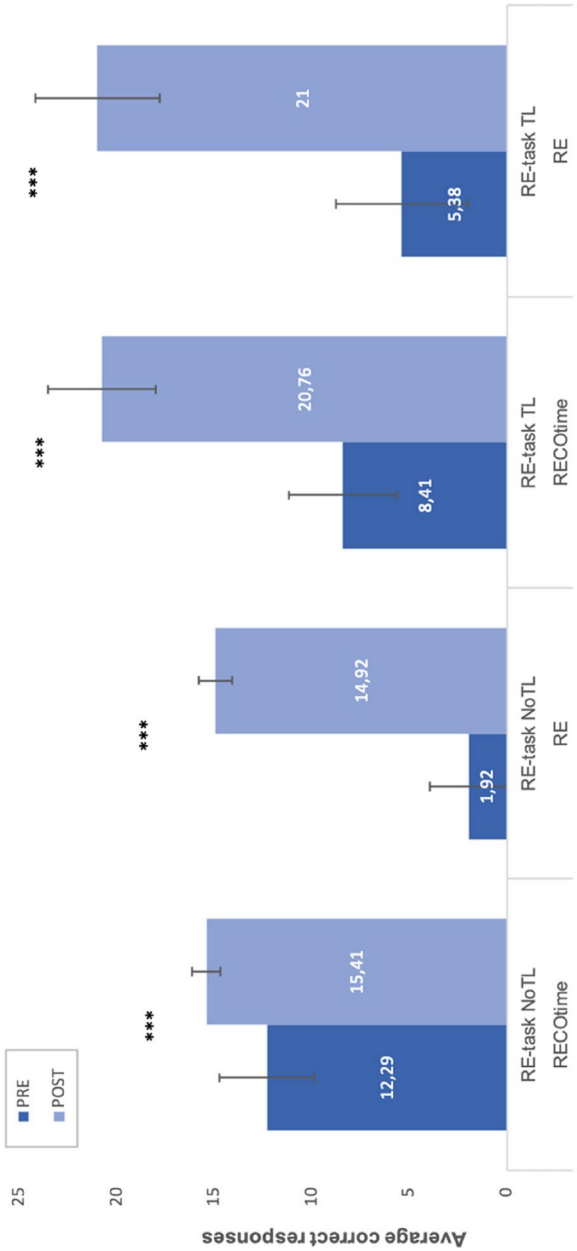


Fig. 3 Average number of correct responses in comparative word problem-solving tasks, pre- and post-intervention, across groups (RE-COtime: participants who showed reversal errors only under time constraints; RE: participants who made reversal errors in both tasks) and task format (RE-task TL: task with a time limit; RE-task NoTL: task without a time limit). *Note.* Error bars represent standard deviations. ***$p < .001$ for significant differences between pre- and post-intervention scores. The RE-task NoTL comprises 16 items, and the RE-task TL comprises 24 items. Values represent the mean number of correct responses for each task

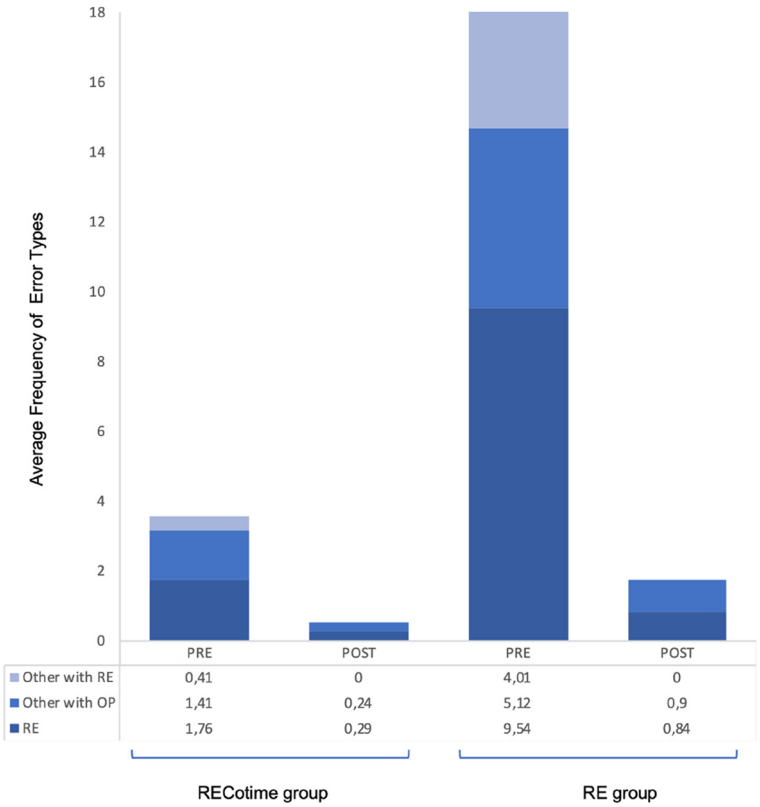


Fig. 4 Average number of algebraic errors, categorized as reversal errors or other error types, by group (RE-COtime: participants who showed reversal errors only under time constraints; RE: participants who made reversal errors in both tasks), before and after the intervention, in the RE-task NoTL. *Note.* Bars represent the mean number of errors in the RE-task NoTL (16 items). Errors are categorized as Reversal Errors (RE) or Other Errors. The *Other Errors* category is further divided into those that also involve reversal error (Other with RE) and those resulting from operator misuse due to linguistic idioms (Other with OP)

Similarly, significant differences in concentration were found between Level 3 and Level 4 ($\chi^2(7) = 1.98$, $p = .048$), and between Level 3 and Level 6 ($\chi^2(7) = 2.64$, $p = .008$), which coincides with a change in the tool's operational mechanics. From Level 4 onward, players received coin-based rewards for accurately translating verbal statements into equations without making reversal errors. This extrinsic reinforcement may have boosted players' concentration levels in later stages due to the presence of immediate feedback and incentive.

Regarding performance-based rewards, the distribution of total coins earned revealed that 50% of participants ($n = 15$) achieved the maximum score of 50 coins. The remaining participants were distributed as follows: 16.7% earned 49 coins, 20% earned 48, 6.7% earned 47, and 3.3% earned 46 or 37 coins, respectively. These results indicate that a large proportion of participants maintained high levels of accuracy throughout the G-IET, with the majority exceeding the threshold required for the economic incentive (≥ 45 coins).

Results from the principal component analysis (PCA) conducted on the average scores of enjoyment, relaxation, and concentration across the eight levels yielded a KMO of.740 and a significant Bartlett's test of sphericity ($\chi^2 = 50.54$, $p < .001$), confirming the suitability of the data for factor analysis. The PCA extracted a single component that accounted for 83.0% of the total variance, with high loadings across all three variables (.900–.931).

This factor was interpreted as a positive emotional engagement dimension, reflecting a latent state characterized by sustained enjoyment, emotional relaxation, and cognitive engagement during interaction with the G-IET. This interpretation aligns with established theoretical frameworks in educational psychology, which conceptualize emotional engagement as the joint presence of positive affect (e.g., enjoyment), attentive absorption (e.g., concentration), and low anxiety or tension (e.g., relaxation) (Csikszentmihalyi, 1990; Fredricks et al., 2004; Pekrun, 2006). Figure 5 presents the average scores of enjoyment, relaxation, and concentration reported at the end of each game level, alongside the score of the first principal component (PCA) derived from these three variables. The figure illustrates how relaxation and concentration increase progressively throughout the game, particularly after the implementation of performance-based rewards at Level 4. The PCA score, which summarizes the shared variance among the emotional measures, shows a similar upward trend. Although this trend is mostly sustained, a slight decrease is observed at Level 8. Nevertheless, the average emotional ratings at Level 8 remain above 4 on the 0–6 scale, indicating that participants continued to experience a moderate to high level of emotional engagement despite the slight decline.

The average flow score, computed as the mean of all five items on a 0–6 scale, was 4.57, indicating a moderately high level of perceived immersion and enjoyment throughout the gamified learning experience. This suggests that the G-IET effectively facilitated a state of flow while supporting the comprehension of comparative word problems and the reduction of reversal errors. Table 1 displays the mean scores and standard deviations for each item.

Correlational analyses revealed a significant positive association between participants' perceived levels of flow and their emotional engagement throughout the educational G-IET ($r = 0.661$, $p < .001$). This finding suggests that higher experiences of flow were associated with increased emotional involvement during interaction with the G-IET.

To examine whether emotional engagement and perceived flow varied by group, it was assessed whether the RE-COtime group, who demonstrated greater initial competence in the RE-task NoTL, exhibited different engagement levels compared to the RE group. A Mann–Whitney U test was conducted to compare the emotional engagement scores (PCA) and the flow score between the RE group and the RE-COtime group. The results revealed no statistically significant differences between the groups (PCA: $U = 94.00$, $p = .489$; flow: $U = 106.5$, $p = .866$), suggesting that participants' reversal error profiles did not affect their levels of emotional engagement or perceived flow during interaction with the G-IET.

3.2 Qualitative insights into the learning experience

The results are presented according to the five open-ended questions posed. The first open-ended question focuses on how the G-IET helped participants understand the difficulties they previously experienced with comparative word problems and their translation into algebraic equations. Thematic analysis of the qualitative responses enabled the identification of four emergent categories reflecting the primary conceptual and strategic obstacles described by the participants. Although the responses primarily refer to difficulties encountered before using the G-IET, the reflective nature of the question revealed increased awareness of the progress made in understanding and solving mathematical word problems following the G-IET intervention (see Table 2).

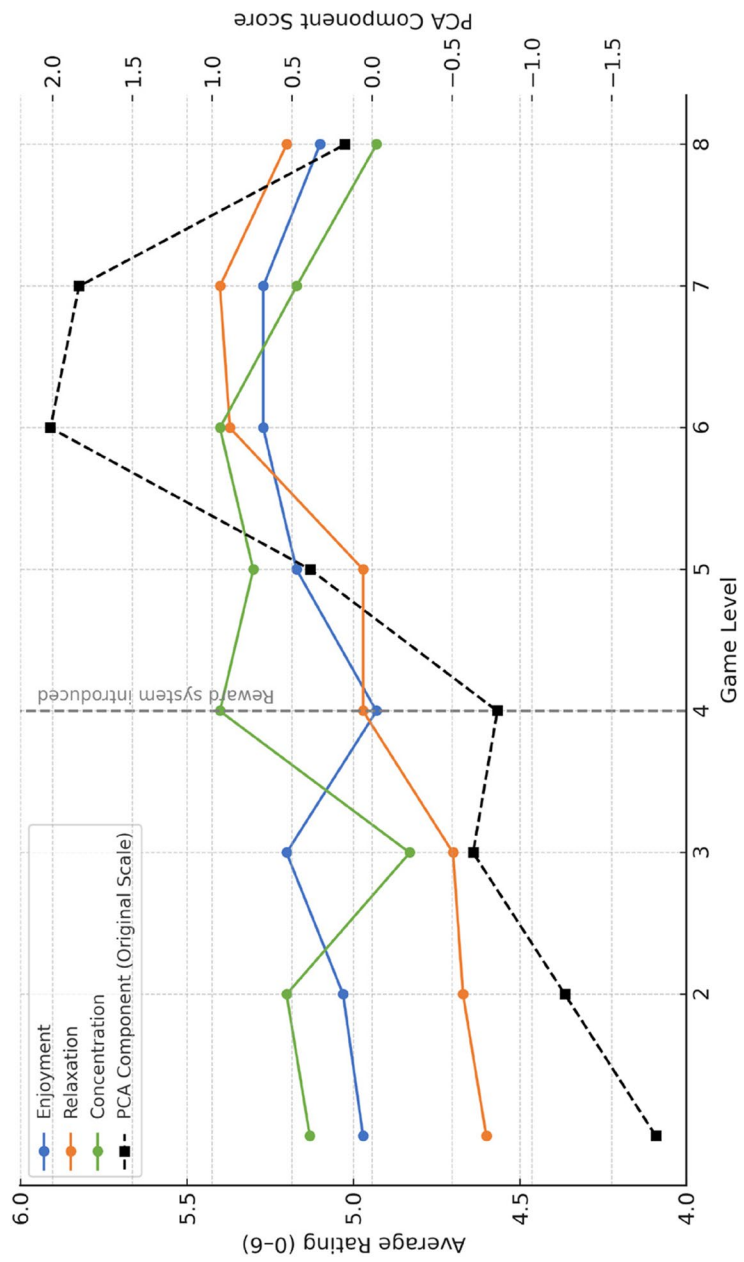


Fig. 5 Evolution of self-reported emotional states (enjoyment, relaxation, and concentration) across the eight levels of the G-IET, alongside the adjusted score of the first principal component (PCA). *Note.* Scores are averaged on a 0–6 Likert scale, with higher values indicating stronger agreement with the positive pole of each dimension (e.g., more fun, more relaxed, more concentrated). The PCA score was inverted to align with the direction of the original emotional ratings. A dashed vertical line marks the introduction of reward-based reinforcement (coin system) starting at level 4, which corresponds to an observable increase in both raw emotion scores and the composite PCA score. The Y-axis is displayed from 4 to 6 (instead of the full 0–6 scale) to improve visual interpretation, since initial ratings were already high and a full-scale plot would obscure relevant trends

Table 1 Mean scores and standard deviations for flow experience items

Item (flow aspect)	Mean	Std. deviation
Time seemed to fly while I was playing	4.73	1.08
I became unaware of my surroundings	4.27	1.86
I felt good while playing	4.80	1.37
I was deeply focused while playing	4.60	1.25
I felt excited while playing the game	4.47	1.31

Table 2 Emerging thematic categories and frequency percentage of common errors made prior to using the G-IET when translating word problems into algebraic language

Emergent category	Frequency (%)	Description
Difficulties in interpreting and translating the problem	37.5%	Literally transcribing the problem without reorganizing the data or analyzing the relationships between quantities
Quick or superficial reading of the word problem	25%	Reading without attention, rushing, or overlooking key details in the problem, which hinders correct analysis
General poor comprehension of the problem	25%	General reading comprehension issues or difficulties interpreting the situation described in the problem
Difficulty identifying correct operators or relationships	12.5%	Errors when choosing between addition/subtraction or multiplication/division; confusion with words like "times", "more", "greater", etc

Several participants expressed more than one idea; therefore, some responses were coded into multiple categories

The most prevalent category, present in 37.5% of responses, was related to difficulties in interpreting and translating the word problem. The responses reveal a tendency to transcribe the problem literally into algebraic language, without reorganizing the elements or properly establishing the relationships between quantities. Some participants stated: "I wrote the equation exactly as the problem said, following it linearly and transforming it into numbers" (P10), or "I think I literally wrote what the problem said, without paying attention to the equality" (P14).

The second most common category, found in 25% of the responses, focused on quick or superficial reading of the problem statement. Participants acknowledged that, before using the G-IET, they tended to read the problems hastily, without taking the time to consider the logical order of quantities or relevant details, which often led to interpretation errors. Following the training provided by the G-IET, they reported having developed greater awareness of the importance of reading carefully, breaking down the problem, and anticipating possible misunderstandings. Representative comments include: "Before, I read the problem quickly and didn't realize which quantity was being compared to which" (P19), and "My main problem was not reading the problem carefully and rushing through it" (P27).

The third category, mentioned in 25% of responses, referred to difficulties in identifying the correct mathematical operators or relationships. Participants indicated that they often selected incorrect operations, such as adding instead of multiplying, or found it challenging to correctly interpret expressions such as "times more", "times less", "more", "greater", or "less." These misinterpretations frequently led to reversal errors in equation

construction. Illustrative examples included: "The symbol I used in the equation was the opposite one" (P20), "Without considering which element is larger or smaller" (P29), "At first, I didn't understand that sometimes it meant multiplying and other times just adding" (P7), and "I also made operator errors and didn't know that 'times' meant multiplication or division" (P6).

Finally, 12.5% of the responses were grouped under the category of general poor understanding of the problem statement. This category includes broader difficulties related to reading comprehension or the global interpretation of the mathematical scenario. A representative example was: "I didn't fully understand the problem statement. I had trouble organizing the data and knowing which of the two was assigned the quantity" (P6).

In response to the second research question, which aimed to explore the strategies and reasoning employed by students prior to engaging with the G-IET, thematic analysis enabled the identification of three emerging categories. These categories reflect the main initial approaches and conceptual errors described by participants when attempting to translate verbal statements into equations (see Table 3).

The most frequent strategy, reported in 50% of the responses, was the literal translation of the problem statement. Participants indicated that, before using the G-IET, they tended to write equations by strictly following the order in which the elements appeared in the text, without reinterpreting their meaning from a mathematical perspective. This approach reveals a superficial reading focused more on the textual sequence than on the underlying logical structure. Representative examples include: "I wrote the equation in that same order" (P2), "The translation of the statement was literal" (P5), "I read the statement as it was and translated it into an equation, nothing more" (P6), and "If they said a truck was five times bigger, I multiplied the truck instead of the other variable" (P8).

Secondly, 37.5% of participants demonstrated a lack of attention to the equality relationship between variables. These responses indicate difficulty in identifying the implicit equivalence relations in the problem statements, which undermines the internal coherence of the formulated equations. As one participant expressed: "I didn't pay attention to equality" (P14).

Finally, 12.5% of responses revealed the use of intuitive or impulsive resolution strategies, where participants attempted to solve the problems based on what "seemed logical", without performing a mathematical verification of the procedure. This kind of

Table 3 Emerging thematic categories and frequency percentages on the strategies previously used for equation construction

Emergent category	Frequency (%)	Description
Literal translation of the problem statement	50%	Direct writing of the equation following the literal order of the problem statement, without semantic reinterpretation
Lack of attention to equality or relationship between quantities	37.5%	Absence of analysis of the equality relationship, which affected the logical coherence of the equation
Intuitive resolution	12,5%	Use of immediate or intuitive strategies, without reflection on the logical structure of the procedure

Several participants expressed more than one idea; therefore, some responses were coded into multiple categories

immediate, unfounded reasoning is illustrated by comments such as: "It seemed logical at first glance, without thinking too much about it" (P20).

In relation to the third research question, which aimed to examine whether the G-IET contributed to an improved understanding of word problem statements and their translation into equations, thematic analysis enabled the identification of five emerging categories. These categories reflect varying levels of awareness, mathematical conceptualization, and strategic adjustment following the gaming experience (see Table 4).

The most frequently observed category (44%) was understanding of equality and equation structure. Several participants reported that, after the intervention, they developed a clearer understanding that comparative word problems represent a relationship of equivalence between two expressions in an equation. They recognized the need to perform appropriate operations to represent the equality between variables. This structural understanding was reinforced by the visual strategy used in the G-IET and the arithmetic support provided through substitution techniques, which guided students in identifying both sides of the equation. One participant noted: "Now I understand that an equation consists of establishing an equality, and that it is necessary to operate on one side so that both sides have the same value" (P8). Other illustrative comments include: "After these sessions, I have developed a better mental representation of the quantities when I read a problem" (P17), "I think the explanations with the drawings especially helped me realize my mistakes" (P12), and "Now I understand the equation in terms of quantities, not in the order of the sentence" (P24).

The second most common category (22.5%) was semantic interpretation and understanding of idiomatic expressions, particularly the phrase "times more/less". Participants reported improved ability to identify the appropriate mathematical operation to apply based on the context of the word problem, learning to distinguish accurately between addition, subtraction, multiplication, and division. This semantic differentiation was considered essential for avoiding common errors. One participant stated: "It helped me mostly to understand when to use addition/subtraction and when to use multiplication/division" (P16).

Table 4 Emerging thematic categories and frequency of occurrence regarding the improvement in the understanding of mathematical word problems and their translation into equations after using the G-IET

Emergent category	Frequency (%)	Description
Understanding of equality and equation structure	44.5%	Improved understanding of the concept of equality and development of an internal logic that enables the identification of which quantity is greater or smaller and how to relate them
Semantic interpretation and idiomatic expressions	22.5%	Correct identification of when to use addition/subtraction or multiplication/division based on the language used in the problem statement
Error recognition and strategic adjustment	21.5%	The G-IET facilitated the identification of previous mistakes and the modification of incorrect strategies
Mechanization and automation of the process	9.1%	Internalization of the steps needed to solve simple equations quickly and automatically
Limited change or no noticeable effect	3%	Participants did not perceive a significant change, although they acknowledged some improvement or increased awareness

Several participants expressed more than one idea; therefore, some responses were coded into multiple categories

Another 21.5% of responses were grouped under error recognition and strategic adjustment. In this category, participants described how, through repetition and the feedback provided by the G-IET, they were able to identify patterns of error in their reasoning and adjust their strategies accordingly. As one student explained: "Through repetitive mistakes, one realizes the need to modify their strategy" (P2).

Additionally, 9.1% of responses reflected a mechanization of the procedure. These participants reported having achieved greater fluency in solving such equations, performing the necessary steps automatically once they had internalized the semantic understanding of the problem statement. A representative comment was: "Now I get it right all the time and have understood what my mistake was" (P27).

Finally, a small group (3%) reported that although the G-IET helped them reflect on their performance, they did not perceive a significant change in how they approached these types of tasks. As one participant expressed: "The G-IET helped me, but it didn't generate much change in how I approach the equation" (P28).

In relation to the fourth research question, which aimed to identify the aspects of the G-IET that participants found most engaging as an educational tool, thematic analysis of the qualitative responses led to the identification of six emerging categories (see Table 5).

The most frequently mentioned category was mathematical understanding and reasoning, present in 44% of responses. Participants indicated that the G-IET helped improve their comprehension of comparative word problems, the structuring of equations, and the application of logical reasoning. Illustrative examples include: "It helped me understand equalities better" (P12), and "I find it a very useful tool for learning and understanding equations" (P5).

A total of 20% of participants referred to a sense of automatization and increased fluency. Their comments suggest that repeated practice helped them internalize and understand the relational structure of the problems, enabling them to quickly recognize the comparative relationship and translate it into an equation with less cognitive

Table 5 Emerging thematic categories and frequency percentage regarding the aspects of the G-IET most appreciated (liked) by the participants

Emergent category	Frequency (%)	Description
Mathematical understanding and reasoning	44%	The G-IET enhances the understanding of mathematical word problems, the proper structuring of equations, and the use of logic in problem-solving. It also reinforces prior knowledge, making its application clearer and more effective
Automatization and improved fluency	20%	The repeated practice offered by the G-IET facilitates the automatization of procedures, increasing speed and accuracy in responses
Attention and active reading	16%	The G-IET improves the ability to read carefully and to identify key structures in problem statements
Confidence and decision-making	14%	The G-IET strengthens confident decision-making and reduces errors caused by impulsiveness
Design and aesthetics	3%	Positive evaluation of the visual design and aesthetic presentation of the G-IET
Playful features	3%	The G-IET is appreciated for its entertaining, dynamic, and enjoyable character

Several participants expressed more than one idea; therefore, some responses were coded into multiple categories

effort. In addition, the tool made them more skilled at identifying which variable was greater or smaller and determining which operator to apply in order to construct an equation that represented the equality between variables. As expressed by some students: "I reached a point where I did it automatically" (P27) and "I feel that the game helped me quickly grasp the mechanics of the questions and consolidate them more easily" (P25).

An additional 16% of responses pointed to enhanced attention and active reading. Participants noted that the G-IET helped them focus more on problem statements, identify key structures, and avoid errors stemming from rushed reading. This is reflected in statements such as: "It helped me pay more attention when reading" (P18), "It helped me to focus more" (P13), and "The G-IET helped me read the problems carefully and reason through them calmly and precisely" (P29).

The category of confidence and decision-making was mentioned by 14% of participants. In these cases, students reported feeling more confident when making mathematical decisions, as reflected in the following comment: "I feel I've gained confidence when deciding on the results" (P8).

Meanwhile, 3% of responses referred to the G-IET's visual design and aesthetics, emphasizing its visual appeal and presentation as positive aspects of the experience. One student summarized this sentiment as: "The game is very aesthetic and visually appealing, really cool" (P9).

Finally, 3% of participants appreciated the G-IET for its playful and engaging character, describing it as fun, dynamic, and enjoyable. A representative example is: "From my point of view, it's an entertaining game…" (P2).

Thematic analysis of the responses to the fifth research question led to the identification of six emerging categories that reflect how participants evaluated the G-IET from a pedagogical perspective. Overall, most participants considered the game to have strong educational potential (see Table 6).

The most frequently mentioned category was applicability in educational settings (31%). Many participants highlighted that the G-IET could be used as a classroom resource, particularly in primary education, to introduce or reinforce content related to equations and equalities. A representative comment was: "As a primary school student teacher, I think it's an excellent tool to apply in the classroom" (P2).

In second place, 24.3% of responses referred to the G-IET's contribution to improving mathematical understanding. Participants emphasized that the G-IET supported better interpretation of basic algebraic concepts, such as equality and quantitative relationships, allowing students to reason more clearly. One participant noted: "It helps you get faster and understand equalities in a more playful way" (P7).

Approximately 18.4% of participants valued the G-IET as a gamified resource that helps sustain motivation and interest, thereby enhancing learning. As one student expressed: "It's dynamic and entertaining… you learn without even noticing" (P12).

In addition, 14% of the responses emphasized the value of learning through repetition and feedback. These participants pointed out that guided practice enabled them to detect errors and automate problem-solving procedures. A particularly illustrative comment stated: "Through repetitive mistakes, you realize you need to change your strategy" (P11).

Table 6 Emerging thematic categories and frequency regarding the usefulness of the G-IET as a pedagogical resource in the educational context

Emergent category	Frequency (%)	Description
Applicability in educational settings	31%	Considered a useful resource for introducing or reinforcing mathematical content in the classroom, with particular relevance in primary education
Improvement in mathematical understanding	24.3%	Supports the comprehension of key concepts such as equality, basic operations, and the structure of algebraic equations
Gamification resource	18.4%	The G-IET's playful elements, appealing visual design, and interactivity promote student interest, motivation, and active engagement
Effectiveness through repetition and feedback	14%	Repeated practice with immediate feedback contributes to the automatization of procedures and the progressive correction of errors
Personal evaluation of learning	7.3%	Individual learning improvements are highlighted, with explicit mentions of perceived personal benefits
Useful under certain conditions	5%	Its usefulness is acknowledged in specific contexts, with certain student profiles, or when applied for limited periods

Several participants expressed more than one idea; therefore, some responses were coded into multiple categories

On the other hand, 7.3% of the responses focused on personal learning experiences, recognizing that the G-IET helped them improve their own understanding or performance, although they did not generalize its usefulness to broader educational contexts.

Finally, while most evaluations were positive, 5% of responses introduced conditional or critical perspectives. These participants suggested that "The video game is useful in structured contexts, but may not be engaging enough when compared to commercial entertainment games" (P30).

4 Discussion

The present study aimed to evaluate a G-IET designed to eliminate reversal errors through a visual learning approach, focusing on the relational understanding of equality in comparative word problems and their translation into algebraic equations. This process was reinforced through strategies of visual representation and arithmetic validation. Quantitative and qualitative results provide converging evidence of the G-IET's didactic value by facilitating substantial improvement in both algebraic performance and students' emotional engagement.

Participants who initially exhibited reversal errors (RE group and RE-COtime group) showed substantial learning gains after interacting with the G-IET, with very large effect sizes in both timed and untimed tasks. This improvement reflects the effectiveness of a gamified environment that provides immediate feedback and visual representation of semantic relationships, serving as a powerful cognitive scaffold (Debrenti, 2024; Sharmin et al., 2024). Furthermore, a near-complete reduction of reversal errors was observed, along with a notable decrease in operator errors related to the idioms "times more/less". The reduction of "hybrid" errors, those involving reversal errors combined with other types of conceptual misunderstandings, further illustrates the G-IET's capacity

to promote deep conceptual restructuring beyond mechanical practice. These findings align with prior research highlighting the potential of digital learning environments to support mathematics learning through interactive activities, immediate feedback, and a motivational, hands-on design (Moyer-Packenham et al., 2008; Pan & Ke, 2023; Steen et al., 2006; Suh et al., 2005).

Emotional outcomes reflected a positive and sustained learning experience. Perceived flow scores at the end of the G-IET suggest a strong sense of immersion and enjoyment among students, highlighting the G-IET's potential to promote engaging learning experiences. At the same time, a factor analysis of enjoyment, relaxation, and concentration revealed that these emotional aspects are closely interrelated and can be conceptualized as a general state of emotional engagement, which progressively increased throughout the game levels. This is consistent with flow theory (Csikszentmihalyi, 1990), which suggests that optimal immersion states, where challenge and skill are balanced, enhance motivation, concentration, and persistence needed to overcome conceptual errors such as reversal error. These findings are also supported by prior studies emphasizing the potential of game-based learning environments to enhance both cognitive and emotional dimensions of mathematics learning (Hamari et al., 2016; Plass et al., 2016). Moreover, integrating mechanics that induce flow may not only sustain attention but also facilitate deeper processing of algebraic structures, supporting more lasting conceptual change.

Although enjoyment remained stable throughout the interaction with the G-IET, relaxation and concentration were found to vary in response to the cognitive demands and level of challenge as students advanced through the levels. Specifically, the increase in these variables after the introduction of rewards (coins) in Level 4 suggests that extrinsic motivators can enhance engagement when aligned with learning objectives (Domínguez et al., 2013). This dynamic illustrates how the tool's design adjusts challenge levels to maintain interest, although a slight drop was observed in Level 8, possibly due to cognitive fatigue, decreased novelty, or anticipation of task completion (Ackerman, 2011).

The positive correlation between perceived flow at the end of the game and sustained emotional engagement across levels supports the link between immersion and emotional motivation (Lee & Hammer, 2011). This association is educationally relevant, as emotional engagement is directly related to intrinsic motivation and persistence in complex tasks (Fredricks et al., 2004; Pekrun, 2006). In the context of this study, the gamified environment not only promoted the cognitive development of algebraic thinking to eliminate reversal errors but also sustained an emotionally stimulating setting, which may have contributed to deeper processing and greater readiness to learn (Hamari et al., 2016; Plass et al., 2016).

Furthermore, the absence of significant differences in emotional engagement and perceived flow between groups with varying initial competence (RE vs. RE-COtime) suggests that the G-IET successfully maintained student interest regardless of prior performance, reinforcing its applicability in heterogeneous classroom contexts.

Qualitative results from open-ended responses provide deeper insights into the evolution of strategies for solving comparative word problems. Prior to the intervention, students demonstrated a surface-level approach: literal translation of problem statements, quick or incomplete reading, reliance on unfounded intuitive strategies, and difficulty in

relating verbal expressions like "N times as many as" to their corresponding operators. These difficulties permeated the entire problem-solving process, from the initial understanding of the statement to its algebraic formalization, highlighting widespread gaps in both semantic and structural interpretation. These limitations align with research indicating students' challenges in translating natural language into algebraic language without explicit instruction in modeling (Castro et al., 2022; Nathan & Koedinger, 2000). According to Nathan et al., (2010), representational fluency, defined as the ability to reason across different symbolic systems and shift flexibly between them, is a key mathematical competency, and one emphasized by the Programme for International Student Assessment (PISA). Moreover, various studies have shown that reliance on syntactic strategies often leads to literal translations of problem statements, without appropriate semantic reformulation (MacGregor & Stacey, 1993; Molina, 2014).

Additionally, it was found that reversal errors often coexisted with other linguistic misunderstandings. Consistent with Jankvist and Niss (2021), the idioms "N times as many as" associated with multiplication/division operators appear to hinder comprehension of comparative word problems. This observation is supported by functional neuroimaging studies, which indicate greater brain activation in regions linked to processing new semantic information and idiom meanings among students who make reversal errors (Ventura-Campos et al., 2022a).

Post-intervention, a positive evolution in problem comprehension, equation formulation, and self-regulation was observed. Students reported better understanding of problem statements and their translation into algebraic equations, demonstrating greater awareness of the explicit equality relationship, improved identification of mathematical operators, and enhanced ability to detect and correct errors. This progression suggests an internalization of new cognitive strategies, facilitated by the immediate feedback and structured progression embedded in the tool's design. These findings align with studies that emphasize the potential of educational games to develop symbolic and semantic competencies in mathematics (Christopoulos et al., 2024; Robles & Quintero, 2020), promoting more concrete and accessible learning experiences. This progress was accompanied by gains in fluency and confidence, a transition described by Oakhill et al., (2003) as indicative of operational integration of knowledge. These improvements are linked to the repetitive practice with feedback, a core design feature of the G-IET, recognized for its effectiveness in consolidating procedural skills in educational contexts (Christopoulos et al., 2024; Hussein et al., 2022; Marazza-Quispe et al., 2024).

From the students' perspective, the G-IET was positively received as a pedagogical resource. They particularly appreciated its visual nature, progressive structure, opportunity to learn without penalty, and the motivation provided by the playful environment. This perception aligns with research emphasizing the motivational value of G-IET, educational video games or serious games when thoughtfully integrated into classroom settings and their meaningful connection to learning outcomes (Yu et al., 2020).

A particularly noteworthy finding is the metacognitive function the G-IET assumed for many students. Beyond serving as a practice tool, it acted as a catalyst for reflection on errors, enabling students to identify their difficulties and adjust their strategies. This ability to reflect on one's performance represents a key dimension of self-regulated

learning (Zimmerman, 2002) and demonstrates the potential of gamified digital environments as tools for developing both cognitive and metacognitive skills.

From an applied perspective, the results highlight the need for instructional proposals that explicitly integrate components such as active reading of problem statements and enhanced instruction in algebraic thinking toward logical reasoning about variable relationships. The diversity of errors and strategies observed pre-intervention suggests that these issues are common among students and that resources like this G-IET can be effective in addressing them.

Overall, the study results demonstrate that the evaluated G-IET not only helps reduce reversal error and other conceptual algebraic errors but also promotes sustained emotional engagement, improved problem-solving fluency, and the development of self-regulation strategies. These findings underscore the importance of designing instructional interventions that simultaneously address cognitive, emotional, and motivational aspects of mathematics learning (Lee & Hammer, 2011), with particular attention to balancing challenge, visual support, and motivation. Designing experiences that promote flow may be an effective pathway to enhancing comprehensive student engagement, especially in subjects like mathematics, where negative attitudes and anxiety often hinder learning (Author), ultimately supporting a deeper and more enduring understanding of algebra.

5 Conclusion

In conclusion, adopting a gamified learning approach to target structural errors in algebra demonstrates clear cognitive and emotional benefits. The intervention supported accurate equation construction by enhancing students' understanding of relational language and symbolic representation. Simultaneously, the G-IET maintained high levels of emotional engagement, reflected in sustained states of flow, enjoyment, relaxation, and concentration throughout the experience.

These findings reinforce existing evidence on the value of well-designed educational games in correcting specific misconceptions in mathematics (Demedts et al., 2024; Garneli et al., 2019), and emphasize the need to simultaneously address cognitive, emotional, and metacognitive components of learning (Lee & Hammer, 2011). In particular, the identification of an integrated latent emotional engagement factor suggests that gamified learning not only activates motivation but also creates a psychologically supportive environment for mathematics learning, where students feel both motivated and cognitively prepared to tackle content-related challenges.

From an educational perspective, this convergence of conceptual precision and emotional engagement offers a promising pathway for designing effective interventions in traditionally challenging domains such as algebra. Creating learning experiences where pleasure, emotional calm, and focused attention coexist may be essential to cultivating a more robust, lasting, and motivating understanding of mathematical thinking.

6 Limitations and future research

Nevertheless, this study has several limitations. First, the small and non-probabilistic sample size limits the generalizability of the findings. Second, data collection relied on self-reported responses, which are subject to potential biases such as memory distortion or social desirability.

Although the G-IET includes gamified elements that likely contributed to the emotional responses observed, the design of the study does not allow us to determine the extent to which these motivational features, as opposed to the instructional components, contributed to the overall learning and emotional outcomes.

Further research is needed to expand the sample size and validate the G-IET in authentic secondary school settings. Implementing the intervention in diverse classroom contexts would allow for a more robust evaluation of its effectiveness across varying student populations, instructional environments, and curricular demands. Such validation would not only strengthen the generalizability of the findings but also provide insights into the practical integration of gamified tools within formal educational systems. Additionally, future studies could explore long-term learning outcomes and the sustainability of cognitive and emotional gains observed during interaction with the G-IET.

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Authors' contributions
Conceptualization: Noelia Ventura-Campos, Mercedes Ventura and Zoe Falomir; Methodology: Mercedes Ventura, Noelia Ventura-Campos and Aida Moreno-Rus; Data Curation: Noelia Ventura-Campos, Aida Morenos-Rus, Mercedes Ventura, Dana Stefanescu; Investigation: Noelia Ventura-Campos, Aida Morenos-Rus and Dana Stefanescu; Formal analysis: Noelia Ventura-Campos and Mercedes Ventura; Writing—original draft preparation: Aida Morenos-Rus, Mercedes Ventura and Noelia Ventura-Campos; Writing—review and editing: Noelia Ventura-Campos, Aida Morenos-Rus, Mercedes Ventura, Zoe Falomir and Dana Stefanescu; Funding acquisition: Noelia Ventura-Campos, Mercedes Ventura and Zoe Falomir; Supervision: Noelia Ventura-Campos; Technology advisor: Zoe Falomir. We would like to note that Aida Moreno-Rus and Mercedes Ventura contributed equally to this work and should be considered co–first authors.

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Data availability
Data are available from the authors upon request, but are not publicly accessible.

Declarations

Ethics approval and consent to participate
The procedures used in this investigation were approved by the Ethics Committee of the Universitat Jaume I. Informed consent was obtained from all participants in this study.

Consent for publication
Informed consent was obtained from the participants in this study to publish the results anonymously.

Competing interests
The authors declare no competing interests.

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